A Harmonic-Comb Parametric Estimator for the Cyclic Spectral Alpha Profile

Abstract — The spectral correlation density (SCD) of a cyclostationary signal is routinely summarized by its *alpha profile*, the maximum of the SCD magnitude over spectral frequency at each cycle frequency. Sparse reconstruction methods for the strip spectral correlation analyzer (SSCA) accelerate SCD estimation by exploiting sparsity in the cycle frequency domain, but their accuracy degrades when the underlying cyclic spectrum is only approximately sparse, as is the case for direct-sequence spread-spectrum modulation. It is shown here that the alpha profile itself is a substantially sparser object than the full SCD surface: for a signal with a single fundamental cycle frequency α_0, the profile's support is confined to the harmonic comb $\{m\alpha_0\}$. A parametric estimator is derived that recovers this comb directly from the signal's cyclic autocorrelation, evaluated on a subsampled index set at a small number of lags. The fundamental frequency is located by a combined-lag detection statistic and refined with a harmonic-consistency rule that resolves sub- and super-harmonic ambiguity. It is further shown that the statistic, although evaluated from few samples, can be computed exactly at every native frequency bin by a single zero-padded fast Fourier transform per lag, rather than by direct summation over a resolution-matched search grid, reducing the dominant computational cost from one that grows with the product of sample count and grid size to one that is independent of sample count. On a direct-sequence spread-spectrum binary phase-shift keyed test signal, the resulting estimator recovers the alpha profile from a few percent of the available samples and executes faster than both dense and sparse SCD reconstruction, at the cost of higher profile-magnitude error relative to full reconstruction.

1 Introduction

A signal $x(t)$ is second-order cyclostationary if its autocorrelation is periodic in time rather than constant, a property exhibited by essentially all digitally modulated communication signals. The spectral correlation density (SCD) $S_X^\alpha(f)$ generalizes the power spectral density to this setting, indexed jointly by spectral frequency f and *cycle frequency* α [1]. For a linearly modulated signal the SCD is concentrated at a discrete set of cycle frequencies determined by the symbol or chip rate, which is what makes cyclostationary analysis a standard tool for signal detection and classification.

The strip spectral correlation analyzer (SSCA) is the conventional algorithm for estimating $S_X^\alpha(f)$ from data [2]. It channelizes the input, forms a channel-data product by multiplying each channel's complex demodulate against the conjugated input, and applies a further Fourier transform along time to resolve α. This last transform dominates the cost of the algorithm, since it is applied once per channel over the full record length. Because the resulting cyclic spectrum is sparse in α for most signals of interest, this transform has been replaced with sparse Fourier transform algorithms, giving a sparse SSCA (S3CA) with substantially reduced sample and computational requirements [3, 4].

Sparse reconstruction of this kind assumes that the cyclic spectrum is close to exactly κ-sparse: that all but a bounded number of coefficients are negligible. This assumption is well matched to a clean tone or narrowband carrier, but is violated by direct-sequence spread-spectrum (DSSS) signals, whose pseudo-random chip sequence spreads energy broadly across the spectrum in addition to the narrow cyclic features associated with the underlying symbol rate. The resulting cyclic spectrum is only *approximately* sparse: a bounded set of coefficients captures the great majority, but not all, of its energy.

This paper considers a different summary of the SCD, the *alpha profile*

$$P(\alpha) = \max_f |S_X^\alpha(f)|, \tag{1}$$

which reports cyclic feature strength independently of spectral location and is the quantity most commonly consulted for detection and classification. It is shown that, unlike the full SCD surface, the alpha profile of a signal with a single fundamental cycle frequency is supported almost entirely on the harmonic comb $\{m\alpha_0 : m \in \mathbb{Z}\}$, and is therefore genuinely sparse even when the underlying SCD is not. This motivates a parametric estimator that recovers the profile directly from a small subsample of the signal, bypassing SCD reconstruction entirely. The remainder of the paper derives this estimator, analyzes its computational cost relative to dense and sparse SCD reconstruction, and reports its accuracy on a DSSS binary phase-shift-keyed (BPSK) test signal.

2 Signal Model and Problem Formulation

The channelized complex demodulate at channel center frequency $f_k = k/N_P$, $k \in \{-\lfloor N_P/2 \rfloor, \dots\}$, is formed as

$$X_T(t, f_k) = \left[\sum_r a(r)\, x(t+r)\, e^{-i2\pi f_k r}\right] e^{-i2\pi f_k t}, \tag{2}$$

with $a(r)$ a length-N_P tapering window. The channel-data product (CDP) is $X_g(t,k) = X_T(t, f_k)\, x^*(t)\, g(t)$ for outer window g, and the SCD estimate follows from a Fourier transform of $X_g(\cdot, k)$ along t: writing $\widehat{X}_g(q,k)$ for its length-N discrete Fourier transform,

$$S_X^\alpha(f_k) = \widehat{X}_g(q,k), \qquad \alpha = f_k + q/N, \tag{3}$$

so that cycle frequency α corresponds to outer-transform bin $q = N(\alpha - f_k)$ for channel k: because the channelizer in (2) already applies a down-conversion by f_k, a *global* cycle frequency α maps to a *channel-relative* transform frequency $\alpha - f_k$, not to α itself. This offset is immaterial for a dense computation, in which the transform is evaluated at every q regardless, but is essential to track correctly in any method that evaluates (3) at a chosen α directly.

Dense evaluation of (2)–(3) over all channels costs $O(NN_P \log N_P)$ for the channelizer and $O(NN_P \log N)$ for the outer transform. Sparse reconstruction replaces the latter with a sparse Fourier transform of the length-N CDP columns, reducing sample and time complexity to a function of the target sparsity κ rather than N [3], at the accuracy cost described above when the true spectrum is only approximately κ-sparse.

3 Harmonic-Comb Profile Estimation

3.1 Harmonic structure

For a signal with a single fundamental cycle frequency α_0 (e.g. a symbol or chip rate), the alpha profile is modeled as supported on

$$\mathscr{A} = \{ m\,\alpha_0 : m = -M_h, \ldots, M_h \}, \tag{4}$$

including $m = 0$, which corresponds to ordinary (non-cyclic) power and is typically the dominant term. The estimation problem reduces to finding the single scalar α_0 and the profile value at each of the $2M_h + 1$ comb points, rather than searching for unstructured support over N candidate locations.

3.2 Cyclic autocorrelation from a subsample

Let $\mathscr{T} \subset \{0, \ldots, N-1\}$, $|\mathscr{T}| = M \ll N$, be a subsampled time-index set. For lag τ, the conjugate cyclic autocorrelation at candidate cycle frequency α is estimated as

$$\widehat{R}_x^{\alpha}(\tau) = \frac{1}{M} \sum_{t \in \mathscr{T}} x(t+\tau)\, x^{(*)}(t)\, e^{-i2\pi\alpha t}, \tag{5}$$

where $x^{(*)}(t)$ denotes either $x^*(t)$ or $x(t)$ depending on whether the ordinary or conjugate cyclic statistic is used; for real-valued modulations such as BPSK the conjugate form is typically the stronger of the two. A single lag is a noisy estimator, so lags $\tau \in \mathscr{L} = \{\tau_1, \ldots, \tau_L\}$ are combined into one detection statistic,

$$T(\alpha) = \left(\sum_{\tau \in \mathscr{L}} \left| \widehat{R}_x^{\alpha}(\tau) \right|^2 \right)^{1/2}. \tag{6}$$

Away from a true cycle frequency, (5) behaves as a zero-mean statistic whose magnitude is set by the additive noise and by the broadband component of the signal; at $\alpha = m\alpha_0$ it accumulates coherently, so that $T(\alpha)$ separates cleanly from its own off-comb floor regardless of how much broadband energy is present elsewhere in the spectrum. This is the property that makes the profile estimator insensitive to the approximate sparsity that limits reconstruction-based methods: the statistic in (6) never represents, and is therefore never corrupted by, energy away from the comb.

3.3 Fundamental frequency estimation

The fundamental is estimated by maximizing $T(\alpha)$ over a plausible range $[\alpha_{\text{lo}}, \alpha_{\text{hi}}]$. Because a true cyclic peak has width $\sim 1/N$ in α, comparable to a native discrete-Fourier-transform bin, a candidate that coincides with a strong harmonic $m\alpha_0$ can score higher than α_0 itself under a coarse search, or the true peak can fall between grid points if the search resolution is too coarse. Both failure modes are avoided with a harmonic-consistency rescoring: for each of the n largest local maxima of T, a comb score

$$C(\alpha) = \sum_{m=1}^{H} T(m\alpha) \tag{7}$$

is computed, and the candidate maximizing C, with ties broken toward smaller α, is retained. A true fundamental accumulates statistic across all H of its own harmonics, whereas a spurious candidate at a multiple or sub-multiple of α_0 generally does not.

3.4 Exact evaluation by zero-padded Fourier transform

Direct evaluation of (6) on a grid fine enough to resolve a width-$1/N$ peak requires $O((\alpha_{\text{hi}} - \alpha_{\text{lo}})N)$ candidate frequencies, each costing $O(M)$ by direct summation, for a total of $O(L(\alpha_{\text{hi}} - \alpha_{\text{lo}})NM)$ operations across L lags. Because $\mathscr{T}$ is a subset of the integers $\{0, \ldots, N-1\}$ rather than an arbitrary real-valued sampling pattern, this direct evaluation is unnecessary. Define the zero-padded lag product

$$z_\tau(t) = \begin{cases} x(t+\tau)\, x^{(*)}(t), & t \in \mathscr{T} \\ 0, & t \notin \mathscr{T}, \end{cases} \tag{8}$$

and let $\widehat{Z}_\tau = \text{FFT}[z_\tau]$ be its length-N discrete Fourier transform. Then, for every native bin q,

$$\widehat{Z}_\tau(q) = \sum_{t=0}^{N-1} z_\tau(t)\, e^{-i2\pi qt/N} = \sum_{t \in \mathscr{T}} x(t+\tau)\, x^{(*)}(t)\, e^{-i2\pi qt/N}, \tag{9}$$

which is exactly $M\widehat{R}_x^{q/N}(\tau)$ from (5), obtained without approximation for every q simultaneously by a single fast Fourier transform. This is not an approximate non-uniform transform requiring interpolation or a spreading kernel [5]; it is exact, because the samples already lie on the native grid and only the transform, not the sampling, is non-uniformly supported. Computing $T(\alpha)$ at all N native bins therefore costs $O(LN \log N)$, independent of the sample count M, in place of the $O(L(\alpha_{\text{hi}} - \alpha_{\text{lo}})NM)$ cost of direct evaluation. A final local refinement over a window of a few bins accounts for α_0 generally not lying exactly on the native grid.

3.5 Profile evaluation on the comb

Given an estimate $\hat{\alpha}_0$, the profile value at harmonic m is obtained from (3), evaluated at the channel-relative frequency $m\hat{\alpha}_0 - f_k$ for each channel k using the same subsample $\mathscr{T}$ and the windowed CDP $X_g(t,k)$:

$$\widehat{S}(f_k, m\hat{\alpha}_0) = \frac{N}{M} \sum_{t \in \mathscr{T}} X_g(t,k)\, e^{-i2\pi(m\hat{\alpha}_0 - f_k)t}, \tag{10}$$

$$\widehat{P}(m\hat{\alpha}_0) = \max_k \left| \widehat{S}(f_k, m\hat{\alpha}_0) \right|. \tag{11}$$

Evaluating (11) for $m = -M_h, \ldots, M_h$ across N_P channels costs $O(M_h N_P M)$, independent of N, since only a bounded number of specific frequencies are queried rather than a dense grid.

Table 1: Complexity of alpha-profile estimation.

Method	Sample cost	Compute cost
Dense SSCA	$O(N)$	$O(NN_P(\log N_P + \log N))$
Sparse (S3CA)	sublinear*	$O(N_P\,\mathrm{polylog}(N,\kappa))$
Comb, direct scan	$O(M)$	$O(L(\alpha_{\mathrm{hi}} - \alpha_{\mathrm{lo}})NM)$
Comb, exact FFT	$O(M)$	$O(LN\log N)$

* for exactly κ-sparse spectra; the channelizer footprint approaches $O(N)$ in practice for the approximately sparse case studied here.

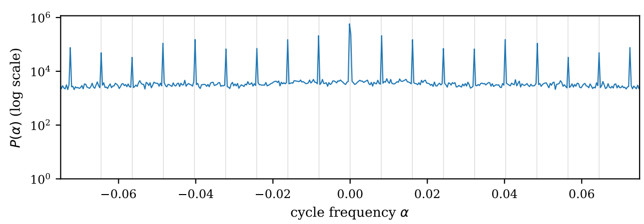

Figure 1: Dense alpha profile of the DSSS-BPSK test signal. Vertical grid lines mark integer multiples of α_0; all significant peaks coincide with them.

Table 1 summarizes the resulting complexity. The direct-scan comb estimator reads far fewer samples than either reconstruction method but is compute-bound rather than sample-bound, since its grid size grows with N for a fixed absolute search width. The exact-FFT evaluation removes this dependence on sample count entirely, leaving a cost governed by N alone and, notably, independent of N_P, since fundamental-frequency estimation does not require resolving spectral frequency and therefore performs no channelization.

4 Experimental Results

The estimator is evaluated on a DSSS-BPSK signal matching the configuration used to validate the original S3CA algorithm [3]: chip rate 0.25, spreading gain 31, and SNR of 10 dB, so that the fundamental cycle frequency is $\alpha_0 = 0.25/31 \approx 8.065 \times 10^{-3}$. Unless stated otherwise, $N = 131072$, $N_P = 32$, and comparisons are made against a dense SSCA reference and against sparse reconstruction using a randomized-hashing sparse Fourier transform [4] at target sparsity $\kappa = 50$ per channel.

4.1 Profile sparsity

Figure 1 shows the dense alpha profile of the test signal on a logarithmic scale. Only 2.1% of cycle-frequency bins carry non-negligible energy, and every populated bin coincides with an integer multiple of α_0, confirming the harmonic-comb model of (4) even though the underlying SCD surface itself is only approximately sparse.

4.2 Fundamental frequency estimation

Figure 2 shows the detection statistic (6), computed exactly at every native bin via (9) and restricted to the search interval, together with the true α_0. The peak is isolated and unambiguous against the surrounding floor. Estimated over 4096 subsampled points, $\hat{\alpha}_0$ matches the true fundamental to within

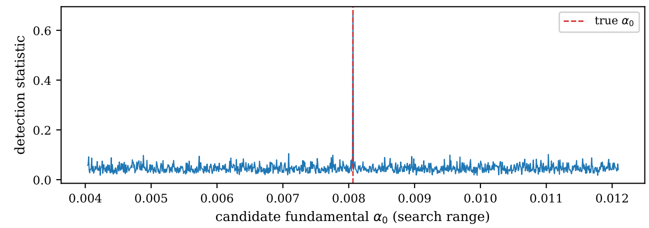

Figure 2: Detection statistic across the fundamental-frequency search range, evaluated exactly via the zero-padded transform of (9).

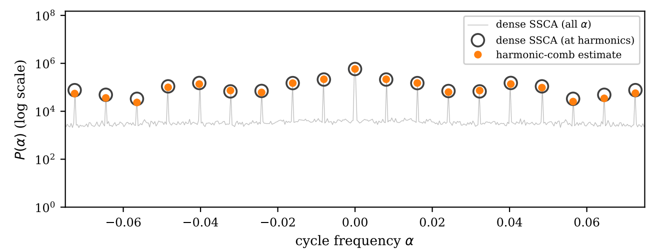

Figure 3: Dense alpha profile compared with the harmonic-comb estimate, on a logarithmic scale. Open circles mark the dense reference value at each harmonic; filled circles mark the estimate at the same location, so that agreement appears as a filled marker centered inside its open ring. The faint background trace is the full dense profile, shown to confirm that no comb location is missed and that the region between harmonics is indeed silent.

0.003% relative error.

4.3 Recovered profile

Figure 3 compares the harmonic-comb estimate, obtained from 6.25% of the available samples, against the dense reference profile on a logarithmic scale, so that strong and weak harmonics are visible at comparable scale. The estimate lands close to the reference at every populated cycle frequency and is silent everywhere else, consistent with the comb model; the small offsets visible between paired markers at a few harmonics are quantified directly in Fig. 4.

4.4 Accuracy relative to reconstruction

Figure 4 compares, harmonic by harmonic, the relative magnitude error of the proposed estimator against that of full sparse SCD reconstruction, both measured against the dense reference. Reconstruction achieves a mean error of 3.0% on populated harmonics, against 8–12% for the comb estimator, reflecting the latter's reliance on far fewer samples. Error is largest for the weakest, highest-order harmonics in both methods.

4.5 Computational scaling

Figure 5 reports wall-clock time for dense reconstruction, sparse reconstruction, and the exact-FFT comb estimator as N grows from 2^{14} to 2^{20} with the subsample count held fixed at 4096. The comb estimator is faster than both reconstruction

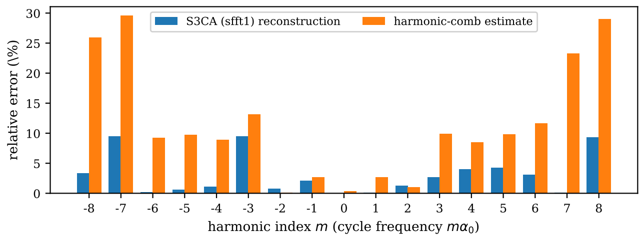

Figure 4: Relative profile-magnitude error by harmonic index, comparing sparse SCD reconstruction and the proposed comb estimator against the dense reference.

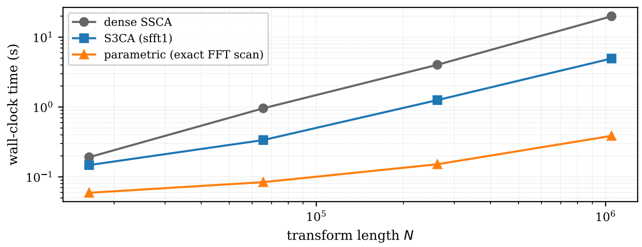

Figure 5: Wall-clock time versus transform length N, fixed subsample count.

methods at every N tested, and its growth with N is markedly gentler, consistent with Table 1: unlike either reconstruction method, it never performs channelization and its cost does not scale with N_P.

5 Discussion

The harmonic-comb estimator trades reconstruction accuracy for sample economy and computational speed. Its accuracy is fundamentally limited by the number of samples used to estimate (5), whereas full reconstruction benefits from a complete, high-resolution transform; the roughly threefold accuracy gap in Section 4.4 reflects this difference rather than any deficiency in the comb model itself, which was shown in Section 4.1 to describe the profile's support exactly. The method assumes a single fundamental cycle frequency; multiple cochannel signals, each with its own baud rate, would require estimating and peeling one comb at a time, analogous to successive interference cancellation. The estimator also returns no information about the SCD away from the comb, so that reconstruction remains necessary whenever the full surface, rather than the profile, is required. Finally, the fundamental-frequency search requires a search range to be supplied; a poorly chosen or excessively wide range increases cost proportionally, since the exact-FFT evaluation covers the full native grid but a wider range trivially admits more false-alarm candidates for the harmonic-consistency rescoring in (7) to reject.

6 Conclusion

The alpha profile of a cyclostationary signal with a single fundamental cycle frequency has been shown to be sparse on the harmonic comb even when the underlying spectral correlation surface is only approximately sparse, and a parametric estimator has been derived that exploits this structure directly, through subsampled cyclic autocorrelation and a harmonic-consistency rule for locating the fundamental. Recognizing that the required non-uniform transform is exact rather than approximate, given integer sample positions, permits its evaluation by a single zero-padded fast Fourier transform per lag, eliminating the dependence of computational cost on sample count and yielding an estimator that is both cheaper in samples and faster in wall-clock time than dense or sparse reconstruction of the full spectral correlation surface, at a moderate cost in profile accuracy.

References

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